Knowledge as the Shared Epistemic Medium

How claims become challengeable, scoped and fit for responsible reliance

SymK

24 August 2026

Contents

Publication status: Coordinated v0.3 Stream H review projection. Human reread, newcomer testing and publication acceptance remain pending.

> **Review status:** Governed v0.3 review draft. The coordinated reconstruction is complete, but Human reread, newcomer testing and publication acceptance remain pending.

Purpose and audience

Purpose. This paper explains why Knowledge is foundational to SymK and why a document, database, model output, consensus or approval is not automatically Knowledge. It is written for an informed newcomer first, while retaining the philosophical and engineering depth required by experienced readers.

Audience. People evaluating, designing, governing or using knowledge-intensive work - including readers with no prior SymK vocabulary. Familiarity with epistemology, artificial intelligence or the SymK evolution history is not assumed.

The question this paper answers. What must be true before SymK may responsibly treat something as Knowledge, or as a claim about Knowledge?

Authority and limits. This is a governed explanatory paper, not a constitutional norm, final semantic package, formal specification or implementation contract. Exact Stage-Accepted decisions control if wording here and those decisions differ.

SymK in plain language

SymK is a general framework and engineering discipline for responsible Knowledge Engineering. It helps people and other capable systems work together on difficult questions while keeping claims, evidence, authority, responsibility, use, consequences and correction visible.

In ordinary language, Knowledge Engineering is the disciplined work of making Domain knowledge expressible, organized, connected to evidence, reviewable, maintainable, challengeable and usable by people and computational systems. It may use vocabularies, ontologies, knowledge bases, models, workflows and AI, but no artifact or technology is the discipline by itself.

Within SymK, Knowledge Engineering is the integrative center and epistemic conditions are its direct engineering object. This paper supplies the concept of Knowledge that the central discipline must preserve: Knowledge is constitutive, not a collateral convenience and not a manufactured output. Other supporting concepts may be admitted later through governed evidence without silently redefining Knowledge or becoming Foundational by implementation use alone.

SymK began with Human-AI cooperation and still treats Domain Intelligence Amplifiers as a principal realization. Its scope is now broader: it can examine governed cooperation among different forms and Bearers of Intelligence. This broader identity does not make every activity part of SymK, erase Human standing or give a technical system authority merely because it is capable.

Why this paper matters. When a team relies on a claim, the important questions are not only whether the words are stored or well presented. We also need to know what the claim concerns, who or what may know it, what supports it, where it applies, what could defeat it, who may authorize reliance, and how correction remains possible.

Two reading paths

Short path - about fifteen minutes. Read this opening, the Abstract, section 1 and the Conclusion. That path states the problem, the central answer, the practical consequence and the limits without requiring the full evolution history.

Full path - for deeper knowledge. Continue through every numbered section, including the counterexamples, distinctions, engineering implications, dissent and references. Those sections support scrutiny and specialist use; they are not prerequisites for understanding the main claim.

The optional *SymK Reader Orientation and Shared Glossary* introduces recurring terms across all four Foundation Papers. This paper nevertheless defines the terms required for its own main argument.

One example to carry through the paper

A bridge-inspection team receives sensor readings, photographs, an engineer's assessment and an AI-generated warning about a crack. The warning is a useful representation, but it is not made Knowledge by appearing in a report. The team must distinguish the physical condition of the bridge, the claim about that condition, the evidence supporting the claim, the proposed Bearer of Knowledge, the scope of the inspection, and the separate decision about whether operational authorities may rely on it. A later repair does not retroactively prove every earlier claim correct.

The example is illustrative. It does not create a universal workflow, legal conclusion or operational rule.

Essential terms for this paper

- **Knowledge:** a scoped epistemic achievement attributable to a Bearer; not merely information that has been stored or accepted.
- **Bearer:** the identified subject or organized system to which a capacity, Knowledge or another property is attributed.
- **Scope and Context:** the boundary and circumstances within which a claim or attribution is meant to hold.
- **Grounds and Evidence:** what supports inquiry or assessment; Evidence bears on claims but does not automatically rewrite an authority or artifact.
- **Representation:** an expression of something for a purpose; the representation is not identical to what it represents.
- **Reliance Authorization:** a separate decision that permits or governs use; it is neither truth nor Knowledge.

Capitalization marks terms that have a controlled SymK use. These short explanations support reading; the Stage-Accepted decisions remain the controlling conceptual source.

Abstract

SymK is a general framework and engineering discipline for responsible Knowledge Engineering through governed cooperation among different forms and Bearers of Intelligence. In that undertaking, Knowledge is the medium through which participants can preserve inquiry, compare claims, coordinate contributions, challenge what is asserted and learn across time. Calling Knowledge a shared medium does not mean that every shared object is Knowledge.

The central problem is simple: storing or communicating a statement does not make it true, known or safe to rely on. A model may produce a correct answer for the wrong reason. A database may preserve an outdated assertion. An authority may permit reliance without establishing truth. SymK therefore distinguishes reality, Knowledge, claims about Knowledge, Grounds, Evidence, representations, assessments, governance status, authorized use and consequences.

The Stage-Accepted minimum is Bearer-relative and non-reductive:

> **Knowledge is a scoped epistemic achievement attributable to a Bearer, in which success concerning what is the case, how to act, or a subject of familiarity is appropriately connected to truth, competence, or encounter rather than to mere luck, assertion, acceptance, or Representation.**

SymK does not claim to settle every philosophical dispute about Knowledge. Its engineering task is to preserve the distinctions and conditions needed for responsible inquiry: who or what the proposed Bearer is, what kind of Knowledge is claimed, where it applies, what supports it, what could defeat it, how it is represented, who may authorize reliance, and how challenge and correction remain possible. The detailed sections develop that foundation, the three coordinated Views, a non-container review envelope and a recursive lifecycle.

Review status and source control

Status: Governed v0.3 Markdown review draft - conceptual reconstruction, cross-paper integration and Stream H Human-review corrections consolidated; Human reread and publication acceptance pending
Version: 0.3 coordinated Human-review draft
Date: 24 August 2026
Source lineage: The v0.1 publications and v0.2 review artifacts remain preserved predecessor objects
Conceptual baseline: Stage-Accepted `SYMK-2X-DR-009` through `SYMK-2X-DR-018`; `2.1-rc.1`
Review lineage: Stream B `SYMK-2X-EV-162`; Stream F cross-paper integration `SYMK-2X-EV-166`; Stream H Human-reading corrections `SYMK-2X-EV-173` through `SYMK-2X-EV-177`
Publication state: Coordinated v0.3 Markdown and PDF review objects produced; complete Human reread, genuine-newcomer testing, any assistive-technology evaluation actually used, and publication acceptance remain pending
Authority boundary: Governed explanatory source; not a constitutional norm, final vocabulary or contract package, formal specification, authority allocation or implementation

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1. How to read this paper

This is a Foundation Paper in the epistemic and human-communication sense. It explains the accepted conceptual baseline, its rationale, its limits and its engineering consequences. It does not acquire constitutional force because it is published in a Foundation directory, uses precise language or receives project-steward review.

SymK's identity is layered:

- its historical Human–AI symbiotic-cooperation protocol remains part of its lineage;
- Domain Intelligence Amplifiers remain its principal realization orientation; and
- its broader identity is the engineering discipline and general framework for responsible Knowledge Engineering through governed cooperation among heterogeneous forms and Bearers of Intelligence.

DIA is principal but non-exhaustive. SymK does not claim all Knowledge Engineering, Intelligence, inquiry, science, ethics, law, organization, education or social cooperation.

Within that identity, Knowledge is foundational because the undertaking cannot become cumulative merely by producing outputs. Participants need to preserve what was asked, what was claimed, why it was considered, which Grounds and Evidence mattered, what Scope and Context bounded it, what alternatives or defeaters remained, which decisions authorized reliance, what consequences followed and what should be revised.

The historical phrase "Knowledge is the shared medium" remains useful if *medium* is not treated as a container. The phrase means that epistemic achievements and their challengeable Representations can mediate inquiry and coordinated action across changes in participant, time and form. It does not mean that every shared object is Knowledge.

Cooperation is a distinct, thin relational process—not the final objective, moral sufficiency, consent, legitimacy, Value or favorable outcome. The accepted SymK value chain is branching, defeasible, recursive and non-automatic. Domain situations may be framed as Questions; inquiry, Evidence acquisition and Reasoning may contribute to Answers; Answers may be assessed for multiple qualities, time, total cost and plural Value; recommendations may inform authorized decisions and actions; consequences may be observed and challenged; and feedback, response, Correction, Learning or Formation may return to any earlier object.

No downstream state retroactively manufactures an upstream predicate. A favorable outcome does not prove that the preceding claim was Knowledge. An accepted claim does not become true because it was accepted. A successful project does not amend universal SymK meaning.

2. Why Knowledge is foundational

A dialogue may produce insight but remain transient. A graph may preserve assertions but omit their Grounds. A model may produce useful outputs but conceal the process by which they were generated. A procedure may be clear while no participant has the practical Knowledge required to perform it. A formally valid conclusion may be irrelevant outside the premises, semantics and Scope of the formal system.

Cumulative inquiry therefore requires more than storage. A participant should be able to recover, proportionately to consequence:

1. the Identity and version of the subject and claim;
2. the Question or purpose to which it relates;
3. the proposed Bearer and the mode of Knowledge attributed;
4. the Domain, Context, Scope, Applicability and time;
5. Grounds, Evidence, counterevidence, Provenance and uncertainty;
6. methods, capabilities, resources, dependencies and failures;
7. assessment, disagreement, challenge and alternatives;
8. governance status and any separate Reliance Authorization;
9. Representations, Projections, transformations and declared loss;
10. participants, affected subjects, Responsibility and competent Authority;
11. use, non-use, consequences, response, Correction and recurrence; and
12. the lineage through which the object was introduced, changed, restricted, superseded or reaffirmed.

Without these distinctions, predictable errors follow:

- a stored statement is treated as true merely because it is present;
- technical validity is treated as epistemic sufficiency outside a formal Scope;
- a Document, graph, database, model or record is mistaken for Knowledge;
- consensus, governance status or Authority is mistaken for truth;
- propositional Knowledge excludes practical Knowledge and familiarity;
- Human or artificial contributions are accepted or rejected solely by origin;
- a cooperative group is assumed to bear every property of its members, or vice versa;
- authorized reliance is mistaken for Knowledge, or Knowledge for permission to rely;
- a readable explanation, scientific model or implementation is allowed to silently rewrite another View; or
- a Domain or project convention silently enters the universal SymK baseline.

SymK's purpose here is not to end philosophical debate. It is to preserve the distinctions and epistemic conditions without which heterogeneous participants cannot responsibly inquire, challenge, decide, act and learn.

3. What the philosophical inquiry contributes

3.1 The tripartite analysis is a useful decomposition, not an inherited settlement

A standard tripartite formulation says that a subject *S* knows a proposition *P* only if *P* is true, *S* believes or accepts *P*, and *S* is justified or warranted in doing so. It is often called the justified-true-belief, or JTB, analysis.

The history is more complicated than a definition transmitted intact from Plato. The *Theaetetus* examines true judgment and the addition of an account without straightforwardly endorsing the modern formula. Contemporary historical analysis also cautions that the idea of a once-universally accepted JTB definition is partly a convenient reconstruction [Ichikawa and Steup 2026].

The decomposition remains useful:

- **truth** separates propositional Knowledge from successful error;

- **belief, acceptance or commitment** identifies a relation to an epistemic subject, although its exact role remains contested; and
- **justification, warrant or support** distinguishes Knowledge from a lucky guess.

These components do not by themselves provide a complete analysis.

3.2 Gettier and the non-accidental connection

Gettier's two 1963 cases present subjects with justified true beliefs that are true through accidental routes [Gettier 1963]. Their force is not that every later theory must use the same vocabulary. It is that correct content plus available reasons can still fail to constitute Knowledge when the connection among claim, Grounds, process and truth is defective.

For Knowledge Engineering, the lesson is immediate:

> **A correct output does not by itself reveal epistemic quality. The relation among claim, Ground, process, competence, alternatives and truth matters.**

A repository may contain a true assertion supported by the wrong reason. A model may answer correctly through a brittle correlation. A rule system may derive a true conclusion from a false premise while another fact makes the conclusion true. Evaluation must therefore preserve process, counterfactuals, defeaters and conditions of failure rather than recording only success.

3.3 Post-Gettier families expose engineering obligations

The literature contains families rather than one settled replacement [Ichikawa and Steup 2026]:

> **Accessible table projection:** Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Family:** defeasibility and no-false-lemmas approaches
  - **Central concern:** relevant falsehoods or undefeated counterevidence
  - **Engineering lesson:** preserve inferential dependencies, defeaters and challenges
- **Family:** reliabilism
  - **Central concern:** truth-conducive process
  - **Engineering lesson:** record method, conditions, performance history and failure distribution
- **Family:** causal approaches
  - **Central concern:** appropriate connection between fact and belief
  - **Engineering lesson:** preserve derivation and production, not only copied origin
- **Family:** sensitivity, safety and relevant alternatives
  - **Central concern:** truth tracking and avoidance of easy error
  - **Engineering lesson:** test boundary conditions and materially nearby failure cases
- **Family:** virtue approaches
  - **Central concern:** success through epistemic competence
  - **Engineering lesson:** represent competence and quality of performance without prestige substitution
- **Family:** contextual and pragmatic approaches
  - **Central concern:** attribution standards may depend on conversational or practical circumstances
  - **Engineering lesson:** declare Context, Scope, stakes, purpose and applicable standards
- **Family:** knowledge-first approaches
  - **Central concern:** Knowledge may be explanatorily fundamental
  - **Engineering lesson:** do not mistake an engineering decomposition for a philosophical reduction

Williamson's knowledge-first program argues that Knowledge can be conceptually fundamental and can help explain evidence, assertion and action [Williamson 2000]. SymK therefore distinguishes two undertakings:

- philosophical analysis asks what Knowledge is and what conditions are necessary or sufficient; and
- engineering characterization asks what must be preserved, tested, governed and challenged when a system supports a Knowledge attribution or Knowledge-related use.

SymK advances the second without claiming to have ended the first.

4. Knowledge has plural modes

The Stage-Accepted baseline recognizes three modes without forcing one mode's criteria or Representation onto another.

4.1 Propositional Knowledge

Propositional Knowledge concerns what is the case. It is truth-apt:

- a medicine has a contraindication;
- a legal deadline expires on a date; or
- a specification requires a stated behavior.

Propositional Knowledge requires truth and an appropriate non-accidental epistemic connection attributable to the Bearer. A statement's presence, formal validity, acceptance, repetition or authorized use does not establish this predicate.

4.2 Practical Knowledge

Practical Knowledge concerns how to act or perform. Ryle made the distinction between knowledge-how and knowledge-that central in analytic philosophy, arguing against the view that intelligent performance always reduces to prior contemplation of propositions [Ryle 1949; Pavese 2022]. The debate continues, so SymK does not require one philosophical reduction.

Its universal minimum is more modest: practical Knowledge requires competence across a relevant task class. A procedure Document is not the competence to perform it. Evidence may include demonstration, repeated performance, transfer across relevant variation, judgment under exceptions, simulation, supervision or accredited assessment. Exact criteria remain Domain, scientific, professional, organizational or project matters.

Practical Knowledge may overlap with Intelligence because both can involve competence across variation. No universal identity or implication is accepted.

4.3 Familiarity Knowledge

Russell distinguished Knowledge of truths from acquaintance with objects and experiences [Russell 1910–11, 1912; Hasan and Fumerton 2024]. The exact philosophical theory remains contested. SymK preserves a bounded minimum: familiarity Knowledge requires a direct or accumulated epistemically significant encounter attributable to a Bearer.

Knowing a person, organization, city, dataset, codebase or practice is not necessarily equivalent to possessing a finite list of propositions. A record of encounters can support an attribution; it does not contain the familiarity or make the attribution true.

4.4 Combination without collapse

The modes may interact:

- propositions can guide practice;
- practice can generate propositions and counterexamples;
- familiarity can supply discriminations important to both; and

- Formation can develop practical Knowledge without reducing Formation to Learning, Training or Education.

Their Bearers, Grounds, Evidence and validation processes may differ. No universal single test, schema or status ladder is adequate to all three.

5. Epistemology and representation ask different questions

Epistemology asks:

- what it is for a Bearer to know;
- whether and how truth, competence, encounter, luck, warrant and commitment matter;
- how disagreement, testimony, collective dependence and Context affect attribution; and
- which defeaters or alternatives undermine a claim.

Knowledge representation and reasoning asks:

- which language expresses selected claims and relations;
- which semantics and constraints govern expressions;
- which conclusions follow under a declared inference regime;
- how uncertainty, inconsistency and identity are handled; and
- what can be computed, stored, queried, transformed or exchanged.

The inquiries meet but do not coincide. A formalism can specify meaning and valid manipulation inside a model. It cannot by itself establish that the premises describe the world, that the Bearer knows, that the model applies, that reliance is authorized or that a consequence is acceptable.

```
subject, practice, situation, or Question
        │
        │ inquiry, encounter, competence, Reasoning, Evidence
        ▼
actual achievement and attribution claims
        │
        │ expression, abstraction, selection, transformation
        ▼
Representations and Projections in one or more Views
        │
        │ inference, retrieval, generation, interpretation, application
        ▼
Answers, recommendations, decisions, actions, outcomes, consequences
        │
        └──────── challenge, feedback, Correction and revision ───────┐
                                                                      │
        ◄─────────────────────────────────────────────────────────────┘
```

Every arrow is typed, scoped, temporal, independently evidenced and challengeable. None is automatic.

6. What representative formalism families provide

The following families are illustrative rather than exhaustive. They are not the universal View inventory and are not selected as SymK's final Ontology, Logic, schema or implementation. Brachman and Levesque (2004) and Baader et al. (2003) provide broad foundations for symbolic knowledge representation and description logics.

6.1 Epistemic modal logic

Epistemic logic uses operators such as `K_a φ`, commonly read as "agent *a* knows that φ," interpreted over possible worlds and accessibility relations. Systems such as S5 are mathematically elegant but place demanding idealizations on agents. Negative introspection and logical closure can conflict with bounded Human and artificial capacities [Rendsvig, Symons, and Wang 2025].

Such logic can represent explicit epistemic commitments and information states under declared semantics. It is not an unqualified psychological model or evidence that an actual Bearer knows.

6.2 First-order logic and rules

First-order logic represents predicates, relations, quantification and implication with model-theoretic precision. Horn-clause fragments support executable rule inference. Classical consequence is normally monotonic inside a fixed theory, while epistemic processes often require restriction, defeat, supersession and return after new Evidence.

SymK therefore separates logical validity from epistemic sufficiency, Applicability and Reliance Authorization. A derived claim preserves premises, assumptions, semantics, inference regime, Scope and known loss.

6.3 Semantic memory networks and frames

Quillian’s “Semantic Memory” used network structure in an early model of semantic information [Quillian 1968]. Minsky’s frames organized stereotyped situations through structured slots, defaults, expectations and alternatives [Minsky 1974/1975].

Their lasting lesson is that epistemic work is not exhausted by isolated propositions. Relations, structured situations, defaults and failed expectations matter. Their historical forms do not become a universal SymK data model.

6.4 Description logics and OWL

Description logics balance expressive power with computational properties. OWL 2 supplies standardized ontology languages; its Direct Semantics is compatible with the model-theoretic semantics of SROIQ, while profiles trade expressiveness for useful reasoning properties [W3C 2012].

An ontology can define selected terms and support valid inferences without certifying that its assertions are true, complete, applicable or responsibly usable.

6.5 Ontologies, RDF and knowledge graphs

RDF defines graphs as sets of triples whose terms occupy subject, predicate and object positions [W3C 2014]. Knowledge-graph practice may combine RDF or property graphs, schemas, identity, Context, rules, embeddings, deductive inference and inductive enrichment [Hogan et al. 2021].

The label *knowledge graph* does not erase epistemic distinctions. A graph may contain observations, hypotheses, accepted claims, disputed assertions, outdated states, inferred claims or uncertain identity links. These differences require explicit governance.

6.6 Probabilistic representations

Bayesian networks represent dependencies among random variables under probabilistic assumptions. Markov Logic Networks attach weights to first-order formulas and combine logical with probabilistic graphical structure [Richardson and Domingos 2006].

Probability is not one universal uncertainty. Statistical frequency, model probability, subjective credence, source confidence, evidential incompleteness and semantic ambiguity are different conditions. The represented quantity, method, calibration, Scope and consequence must remain visible.

6.7 Neural representations

Neural models encode learned regularities across distributed parameters rather than only explicit symbolic assertions. A 2019 study showed that then-current pretrained language models could recall some relational facts and examined whether they might serve some knowledge-base functions [Petroni et al. 2019]. That result does not establish that every later model stores Knowledge or that generated output is self-authenticating.

Neural systems may support discovery, retrieval, synthesis, pattern recognition, hypothesis generation, critique and translation. Their outputs remain Contributions to an epistemic process unless and until the relevant predicates are separately supported.

6.8 Comparative boundary

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Family:** epistemic modal logic
  - **Selected strength:** agent-relative modal reasoning
  - **Typical limitation:** idealized closure and introspection
  - **Appropriate SymK role:** scoped formal Projection of explicit epistemic states
- **Family:** first-order logic
  - **Selected strength:** precise expressive entailment
  - **Typical limitation:** general inference limits and fixed-premise assumptions
  - **Appropriate SymK role:** semantic and deductive Projection
- **Family:** Horn rules
  - **Selected strength:** executable inference
  - **Typical limitation:** restricted expressiveness and policy-sensitive revision
  - **Appropriate SymK role:** operational rule Projection
- **Family:** frames and semantic networks
  - **Selected strength:** structured situations, defaults and relations
  - **Typical limitation:** informal or system-specific semantics
  - **Appropriate SymK role:** explanatory or Engineering Representation
- **Family:** description logics / OWL
  - **Selected strength:** classification under decidable fragments
  - **Typical limitation:** expressiveness/performance and open-world assumptions
  - **Appropriate SymK role:** ontology Projection and validation support
- **Family:** RDF / knowledge graphs
  - **Selected strength:** identity, integration and traversal
  - **Typical limitation:** triples do not carry intrinsic warrant or truth
  - **Appropriate SymK role:** graph Projection of claims and Provenance
- **Family:** probabilistic models
  - **Selected strength:** quantified selected uncertainty
  - **Typical limitation:** model and parameter dispute
  - **Appropriate SymK role:** Scientific or Engineering Projection
- **Family:** neural models
  - **Selected strength:** pattern learning and flexible generation
  - **Typical limitation:** weak claim-level Grounds and correction visibility
  - **Appropriate SymK role:** epistemic participant and representational instrument

SymK does not choose one family and call it Knowledge. Each material Projection identifies its governed subject, source and target View, method, assumptions, additions, omissions, uncertainty, Authority, reversibility and material loss.

7. Distinctions the Knowledge foundation must preserve

7.1 Actual Knowledge and Knowledge attribution

Actual Knowledge is the scoped epistemic achievement. A Knowledge attribution is a fallible claim that a proposed Bearer actually bears that predicate. Recognition, approval or a record does not create actual bearing.

Every material attribution identifies the proposed Bearer, its stable boundary, the Knowledge mode, object, Domain, Context, Scope, time, criteria, Grounds, Evidence, failures, alternatives, uncertainty, assessor, Authority, intended use, status, challenge and lineage.

7.2 Knowledge claim

A Knowledge claim presents epistemic content or attributes Knowledge. It may be true or false, supported or unsupported, accepted or rejected, applicable or inapplicable, authorized or unauthorized for a use. Labeling it a Knowledge claim does not make it Knowledge.

Claim content, assertion occurrence, source commitment, Knowledge attribution and the carrier that represents them remain distinct.

7.3 Ground, Evidence and Provenance

- A **Ground** is a reason, process, competence, experience, encounter or relation materially relevant to a claim or achievement.
- **Evidence** is the contextual role something plays in supporting, challenging or assessing a specified object under declared criteria.
- **Provenance** preserves origin, participation, custody, derivation, transformation and Responsibility.

An object does not possess one intrinsic, universal Evidence status. The same Document or observation may play different roles for different claims. Ground, Evidence and Provenance may contribute to epistemic support; none alone entails truth, Knowledge, warrant, Applicability or Reliance Authorization.

7.4 Epistemic Assessment and governance status

Epistemic Assessment is an act and result of evaluation. Governance status records a process-specific position. Lifecycle status, challenge state, representation integrity, normative validity and legal authority are other distinct conditions.

`Accepted` never means “made true.” A well-supported claim may remain unauthorized for a use because of Scope, risk, privacy, professional Responsibility or external Authority. Reliance may sometimes be authorized proportionately under uncertainty without a claim that actual Knowledge has been achieved.

7.5 Knowledge asset, record and Document

A Knowledge asset is a governed engineering aggregate or Representation supporting preservation and use. An Epistemic record represents a claim, attribution, assessment, decision, relation or event. A Document is a Representation or resource that may play contextual epistemic roles.

None is a metaphysical container of Knowledge. Asset, record or Document status creates no truth, bearing, warrant, Authority or authorization.

7.6 Representation and Projection

A Representation is a situated form through which selected aspects of a subject are made available for interpretation, communication, Reasoning, storage or action. Projection is the governed relationship through which selected dimensions of a source are expressed in another Representation, with transformations, additions, omissions, and known loss declared.

A Projection may add, omit, strengthen, weaken or transform. A reduced operational Projection can refer to a fuller governed record, but it may not present itself as complete when material meaning has been lost.

8. The Stage-Accepted Knowledge minimum

The governing SymK minimum is:

> **Knowledge is a scoped epistemic achievement attributable to a Bearer, in which success concerning what is the case, how to act, or a subject of familiarity is appropriately connected to**

> **truth, competence, or encounter rather than to mere luck, assertion, acceptance, or Representation.**

Each part establishes a boundary.

Scoped

Every material attribution states Domain, Context, Scope, Applicability and time. Scope may include population, task, assumptions, stakes and conditions. Scope does not make truth relative; it makes the attribution and allowed inferences precise.

Actual Scope, declared Scope and authorized-use Scope remain distinguishable. Misstating the Scope can produce a false or misleading attribution even when a claim is useful elsewhere.

Epistemic achievement

Knowledge is not accidental success. *Achievement* leaves room for non-accidental connection, competence and encounter without selecting one total philosophical reduction. Preserved dissent remains: some modes may be better understood as a standing state or relation.

Attributable to a Bearer

Every actual Knowledge instance has a Bearer. An attribution identifies the organized subject proposed to occupy that role and may be mistaken. The Bearer is not automatically the source, Evidence producer, assessor, Authority, owner, responsible party or affected subject.

A participant's Knowledge does not automatically become group Knowledge. A group's artifact does not make the group a literal Knower. Collective, institutional, distributed, artificial and hybrid cases require level-specific Identity, boundary and Evidence.

Success concerning what is the case, how to act, or a subject of familiarity

The three phrases correspond to propositional, practical and familiarity Knowledge. They may combine. No mode is reduced to another's Representation or validation process.

Appropriately connected to truth, competence or encounter

The connection excludes mere luck, assertion, acceptance and Representation. Its exact assessment is mode-, Domain- and consequence-sensitive. SymK preserves the universal minimum and leaves substantive epistemic standards to competent scientific, professional, Domain or external authorities.

Not defined through Intelligence or reliance

Knowledge and Intelligence are distinct Foundational concepts. Neither definitionally presupposes the other, and attribution of one does not automatically establish the other. Their relations are directional, typed, scoped, contingent and challengeable.

Responsible reliance is a consequence-sensitive concern, not a constitutive phrase inside the Knowledge definition. Knowledge does not authorize its own use. Reliance Authorization identifies a competent, scoped governance relation with conditions, safeguards, review and expiry.

9. Human, Scientific and Engineering Views

SymK governs three coordinated semantic Views. They communicate, investigate and realize selected aspects of the same governed subject. They are not three independent bodies of Knowledge and none is sovereign.

9.1 Human View

The Human View preserves meaning, purpose, definitions, lived experience, practical Knowledge, participation, affected subjects, Power, trust, burden, challenge, refusal, response, alternatives, dissent, consequences and known limits.

Human readability or approval does not establish scientific, engineering, legal or constitutional validity.

9.2 Scientific View

The Scientific View preserves Questions, constructs, hypotheses, observations, measurement, methods, Evidence, counterevidence, causal claims, sampling, uncertainty, defeaters, reproducibility, empirical adequacy, generalization and conditions of defeat.

Scientific prestige or measurement does not create universal semantic or normative Authority.

9.3 Engineering View

The Engineering View preserves identities, versions, schemas, constraints, interfaces, states, events, dependencies, Provenance, inspectability, monitoring, validation, failure handling, Correction, revalidation, migration and interoperability.

Formal precision, conformance or operational success does not establish meaning, truth or authorized use.

9.4 Alignment and loss

Human readability, formal expression and machine inspectability are cross-view capabilities. Natural language is not automatically Human View; mathematics is not automatically Scientific View; code is not automatically the complete Engineering View.

Material alignment examines Human–Scientific, Scientific–Engineering and Human–Engineering correspondence. Pairwise agreement does not establish triangular coherence when the third View diverges.

Every material transformation declares source and target View, object and version, method, aggregation, omission, uncertainty, Authority, reversibility and material loss. Evidence may challenge another View or the governed subject, but it does not amend the subject or its authority source automatically. Change follows competent procedure and retains mismatch and correction lineage.

10. A non-container review envelope and recursive lifecycle

v0.1 presented a `KnowledgeObject` tree as a conceptual inventory. That artifact is preserved as history but is not admitted as a Foundational concept, subtype of Knowledge, canonical schema or package anatomy.

For review, the following envelope may be used proportionately. It is a checklist of distinct roles, not one universal object:

```
governed subject and Identity
├── Question, purpose, Domain, Context, Scope, Applicability and time
├── proposed Bearer and Knowledge mode
├── claim content and attribution occurrence
├── Grounds, Evidence, counterevidence, alternatives and uncertainty
├── Provenance, participants, transformations and dependencies
├── Epistemic Assessment, disagreement and challenge
├── governance status and separate Reliance Authorization
├── affected subjects, Responsibility, competent Authority and response routes
├── Human, Scientific and Engineering Representations
├── Projection methods, omissions, uncertainty and semantic loss
└── use, consequences, Correction, revision, supersession and history
```

An implementation may distribute these roles across records, services or artifacts. Their arrangement does not change their meanings. Final package anatomy, identifiers, relation cardinalities, schemas and formal constraints remain for later stages.

10.1 Recursive lifecycle

The epistemic lifecycle is not a promotion ladder:

```
purpose and Question formation
        ↕
participant and boundary identification
```

```
        ↕
inquiry, encounter, acquisition and Formation
        ↕
expression, preservation and transformation
        ↕
assessment, disagreement and challenge
        ↕
possible Reliance Authorization and application
        ↕
outcome and consequence observation
        ↕
response, Correction, revision, restriction, supersession or retirement
        ↕
new Questions, Learning, Formation and institutional memory
```

Any phase may return to any earlier object. Knowledge can precede expression. Practical Formation can precede explicit claim. Familiarity can arise without a discrete proposition. A challenge can enter before or after use. A lifecycle transition changes an activity, Representation, assessment, authorization or operational condition; it does not create truth or actual Knowledge.

11. Explanatory obligations from the accepted baseline

These are not proposed axioms. They explain obligations already traceable to Stage-Accepted DR-009–018 and remain subordinate to the exact accepted decisions.

1. **Knowledge and Representation remain distinct.** No Document, graph, rule, record, prompt, embedding, model weight or package is identical to the Knowledge it represents or supports.
2. **Truth is not created by governance.** Approval, consensus, Authority, repetition, storage and adoption do not make a proposition true.
3. **Every material attribution is scoped.** Domain, Context, Scope, Applicability and time constrain what is claimed and where it applies.
4. **Ground, Evidence and Provenance remain typed.** They support inquiry and assessment without individually entailing truth, Knowledge, warrant or authorized reliance.
5. **Assessment, governance status and Reliance Authorization remain distinct.** No single lifecycle field can substitute for them.
6. **Disagreement and contradiction remain visible before resolution.** Different Scope, vocabulary, method, Authority, time or Evidence may explain an apparent conflict; storage convenience does not justify silent merging.
7. **Knowledge modes remain plural.** Propositional, practical and familiarity Knowledge may combine without one universal Representation or test.
8. **Formal inference is conditional epistemic work.** Premises, semantics, inference regime, assumptions, Scope and known loss remain attached to derived claims.
9. **Origin does not settle epistemic standing.** Human, artificial, institutional and hybrid Contributions are assessed by predicate-specific criteria, not prestige or origin alone.
10. **Agreement is not Knowledge.** Consensus is distinct from truth and warrant; distributed dependencies do not automatically create a collective Bearer.
11. **Challenge and amendment remain distinct.** Evidence may challenge claims, premises, Grounds, assumptions, Applicability, consequences, decisions and exercises of Authority. An authority source or governed artifact changes only through competent procedure.
12. **Implementation does not own meaning.** Storage and inference technologies may test or project governed semantics; they do not silently define them.

12. Heterogeneous participation, Authority and affected standing

Human and artificial systems may retrieve sources, observe, generate hypotheses, formalize, infer, simulate, critique, translate, test and maintain. These can be material Contributions. No participant type is automatically a Knower, Authority or responsible subject for every object.

Epistemic work may distribute roles such as:

- source, observer or encounter participant;
- claimant or proposed Bearer;
- interpreter or translator;
- formalizer or implementer;
- Evidence producer or custodian;
- critic or challenger;
- assessor or validator;
- decision-maker or reliance authority;
- adopter, user or maintainer; and
- affected subject or representative.

Role assignment, occupancy, capability, performance, Authority, Authorization, Responsibility and record remain distinct. Competence does not create permission; Authority does not create truth; contribution does not create whole-result Responsibility; and use does not prove Knowledge.

The accepted baseline distinguishes eleven typed Authority kinds across bounded governance loci. This paper allocates none of them. Any material exercise identifies source, kind, jurisdiction, Domain, Context, Scope, subject, object, time, conditions, challenge and lifecycle.

Affected-subject standing/challenge remains a distinct unresolved locus, not a settled twelfth Authority kind. Materially affected subjects should remain representable and should have access, where applicable, to Evidence contribution, challenge, capable assessment, reasoned response and competent routes. This does not automatically create decision Authority, veto, legal standing, consent, Responsibility or a particular remedy.

Evidence may challenge claims, premises, Applicability, consequences, and decisions operating under any Authority. It does not automatically amend a governed artifact, alter or revoke an Authority or its source, or reverse a decision; such change follows competent procedure. Urgent adverse Evidence triggers urgent consideration by competent operational or external Authority. This creates no emergency duty, containment or suspension power, decision right, remedy, or jurisdiction.

Different participants need not possess identical internal Representations. Cooperation requires sufficient semantic alignment, traceable translation and visible disagreement—not cognitive uniformity or automatic collective Knowledge.

13. Development requirements

Any later SymK realization supporting material Knowledge claims should satisfy the following requirements proportionately to consequence.

13.1 Separate the layers

Distinguish at least:

- subject from Representation;
- actual bearing from attribution;
- claim from truth and Knowledge;
- Ground from Evidence and Provenance;
- assessment from governance status;
- status from Reliance Authorization;
- logical validity from empirical and epistemic sufficiency;
- confidence from probability and other uncertainty kinds;
- source from Authority and Responsibility;

- active record from current applicability; and
- present state from historical state.

13.2 Preserve epistemic lineage

For a material claim, enable governed recovery of:

1. who or what introduced it;
2. what it meant in that version;
3. which sources, processes, competences or encounters supported it;
4. which transformations and participants affected it;
5. which objections, failures and alternatives were recorded;
6. which assessment and status were assigned, by whom and under what Authority;
7. where and why reliance was authorized or refused;
8. which Representations and Views express it;
9. which outcomes and consequences followed; and
10. what would restrict, defeat, supersede or reopen it.

13.3 Support non-monotonic and recursive evolution

New Evidence may restrict, defeat or supersede an earlier position. Preserve prior states for audit and Learning without presenting them as current. Allow challenge, consequence and new inquiry to re-enter at any material point.

13.4 Preserve plural Representations and Three-View alignment

Natural-language explanations, scientific models, formal axioms, graph structures, rules, tests, code and runtime records may express selected aspects of one governed subject. Trace their source, target, version, method, Authority, omissions and loss. A change in one does not silently rewrite the others.

13.5 Type uncertainty and disagreement

Do not reduce all uncertainty to one scalar. Distinguish, where material:

- source confidence;
- probability or frequency;
- incomplete Evidence;
- semantic ambiguity;
- model uncertainty;
- disagreement among participants;
- instability across Contexts;
- unresolved contradiction; and
- uncertainty about Applicability, consequence or authorized use.

13.6 Preserve negative results and affected consequences

Failed hypotheses, counterexamples, rejected formulations, boundary failures, abstentions and adverse consequences may be important epistemic assets when their status and rationale are explicit. Correction does not erase history, residual effects or recurrence risk.

13.7 Prevent premature universalization

A concept validated in a Domain or project is Evidence for SymK, not automatic universal meaning. Expose Domain assumptions and route possible generalization through competent concept governance. No project Ontology, Logic, schema, workflow, metric or implementation gains semantic Authority by success or reuse.

13.8 Preserve change procedure and consequence return

Evidence challenge is not extra-procedural amendment. Record the challenged object, Evidence, assessor, competent Authority, decision, changed artifact, version and migration. Outcomes and consequences may inform revision only through governed Evidence and causal assessment.

14. Accepted answers and live questions

The v0.1 paper posed eight questions. Later Stage-Accepted work answered parts without eliminating dissent or downstream ownership.

14.1 Is propositional Knowledge factive?

Accepted minimum: propositional Knowledge requires truth and an appropriate non-accidental epistemic connection. `KnowledgeClaim`, assessment, governance status and Reliance Authorization represent fallible system positions without weakening factivity.

Live question: exact commitment or belief conditions may vary by Bearer class and remain philosophically contested.

14.2 Can a cooperative system literally know?

Accepted minimum: cooperation does not automatically create a collective Bearer, and participant properties do not transfer to the whole. Every whole-level attribution requires stable Identity, integration, boundary and Evidence at the proposed level.

Live question: which organizations, distributed teams or hybrid configurations satisfy literal Knowledge-bearing criteria remains case-sensitive.

14.3 What qualifies an artificial system as a Knowledge Bearer?

Accepted minimum: origin neither establishes nor disqualifies standing. Attribution must identify a proposed Bearer and mode-specific criteria, Grounds, Evidence, limitations and alternatives. Intelligence, Agency, autonomy, moral standing, legal status, Authority and Responsibility are separate predicates.

Live question: contemporary artificial configurations may lack or satisfy different conditions across modes; product labels do not settle the issue.

14.4 How should practical Knowledge be governed?

Accepted minimum: practical Knowledge requires competence across a relevant task class. Documents and demonstrations may support attribution without containing the competence.

Live question: Domain-specific maturity, transfer, supervision and validation standards remain for competent authorities.

14.5 What is the relation between Knowledge and Understanding?

Accepted minimum: no universal identity or implication has been accepted.

Live question: Understanding may be a Knowledge mode, an Intelligence dimension, both or another concept.

14.6 How should Context affect attribution and reliance?

Accepted minimum: distinguish proposition content, Context, actual Scope, declared Scope, Applicability, assessment standard, stakes and Reliance Authorization. They do not collapse into one `context` field.

Live question: exact epistemic and pragmatic effects remain Domain- and theory-sensitive.

14.7 How can conflicting but locally successful models coexist?

Accepted minimum: preserve model Identity, purpose, scale, Scope, Evidence, assumptions, alternatives, contradictions, explanatory difference and consequences. Local success does not establish universal truth or equivalence.

Live question: Domain sciences own substantive inter-model relations and admissible pluralism.

14.8 What are the irreducible primitives and final package?

Accepted minimum: the integrated Stage 2.1 baseline has seventeen Foundational anchors. Claim, Ground, Evidence, Provenance, assessment, status, reliance, asset, Bearer and Projection have governed supporting or artifact routes.

Live question: final `FC-*` package anatomy, identifiers, relation cardinalities, formal semantics and implementation remain for later stages. This paper does not pre-empt them.

15. A material review envelope

Before relying materially on a Knowledge-related contribution, reviewers may ask the following. This envelope does not prove Knowledge, admit one canonical package or authorize reliance.

> **Accessible table projection:** Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Dimension:** subject and Identity
  - **Review question:** What exact subject, claim, capability or familiarity relation is at issue, in which version?
- **Dimension:** Question and purpose
  - **Review question:** Which Question or purpose does it address, and what would count as responsive?
- **Dimension:** Bearer and mode
  - **Review question:** Which proposed Bearer and which Knowledge mode are claimed?
- **Dimension:** non-accidental connection
  - **Review question:** What truth connection, competence or epistemically significant encounter supports the attribution?
- **Dimension:** Domain and bounds
  - **Review question:** Which Domain, Context, actual/declared Scope, Applicability, time, population, task and assumptions apply?
- **Dimension:** Ground and Evidence
  - **Review question:** Which reasons, processes, competences, experiences or encounters are Grounds, and what Evidence role do particular objects play?
- **Dimension:** alternatives and failure
  - **Review question:** Which defeaters, counterexamples, competing explanations, failed tests and unknowns remain?
- **Dimension:** Provenance and lineage
  - **Review question:** Can origin, participation, custody, derivation, transformation, Responsibility and version history be recovered?
- **Dimension:** assessment and status
  - **Review question:** Which assessment, criteria, assessor, uncertainty, disagreement, challenge and governance status apply?
- **Dimension:** reliance
  - **Review question:** Is reliance separately authorized for this use, by which competent Authority, with which safeguards, review and expiry?

- **Dimension:** Views and Representation
  - **Review question:** Which Human, Scientific and Engineering Representations exist, how were they transformed, and what material loss remains?
- **Dimension:** Authority and standing
  - **Review question:** Who may contribute, assess, decide, challenge and respond; which affected subjects and competent routes remain visible?
- **Dimension:** responsibility and consequence
  - **Review question:** Which Responsibility relations, intended uses, possible outcomes, harms, benefits, plural Value claims and affected distributions matter?
- **Dimension:** revision
  - **Review question:** What Evidence or consequence would restrict, defeat, correct, supersede, retire or reopen the position?
- **Dimension:** broader relation
  - **Review question:** How does the contribution relate to SymK's layered identity, possible Cooperation and the branching Question-to-consequence value chain without claiming automatic improvement?

A contribution may be preserved even when the review exposes uncertainty, disagreement or a negative result. The correct outcome may be continued inquiry, qualified no-answer, restricted use, refusal, supersession or explicit non-admission.

16. Conclusion

The literature does not give SymK one definition or formalism ready to copy. It reveals why Knowledge cannot be equated with stored assertions, approved records, formal validity, fluent output, consensus or successful action.

The accepted SymK minimum is:

> **Knowledge is a scoped epistemic achievement attributable to a Bearer, in which success concerning what is the case, how to act, or a subject of familiarity is appropriately connected to truth, competence, or encounter rather than to mere luck, assertion, acceptance, or Representation.**

Knowledge Engineering does not place this achievement into a file. It designs, constructs, governs, evaluates and evolves epistemic conditions that may enable, constrain, preserve, expose or impair how Knowledge is pursued, achieved, attributed, expressed, communicated, assessed, challenged, applied and revised.

Knowledge remains the shared epistemic medium when its claims, Grounds, Evidence, Provenance, Context, Scope, uncertainty, dissent, Authority, Representations, consequences and history can survive changes in participant and form. The medium metaphor creates no container and no collective Knower.

Cooperation may help participants combine Contributions and capabilities, but it is a process rather than the final objective. Answers, decisions, actions, outcomes, consequences and plural Value remain separately evidenced and challengeable. The system remains open to Correction, Learning, Formation and new Questions.

That is the foundation this paper supplies: not a claim that SymK stores truth, but a disciplined commitment to preserve the conditions under which heterogeneous participants can inquire into, challenge, responsibly use and revise what they claim to know.

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Document governance note

This v0.3 review draft is a new governed edition derived from the preserved v0.2 review state. It does not overwrite v0.1 and is not publication-accepted. Stream F cross-paper integration is complete analytically through `SYMK-2X-EV-166`. Later changes must distinguish:

- correction of source or bibliographic error;
- change to explanatory wording;
- challenge to an attribution or claim;
- proposed change to the Stage-Accepted semantic baseline;
- cross-paper terminology or entailment correction;
- Human-comprehension or accessibility correction;
- publication-only correction; and
- later package, formal or implementation work.

Evidence may challenge this paper's claims, premises, Applicability, consequences and decisions. The paper or its governing authority source changes only through competent procedure with preserved lineage. No implementation may silently promote the review envelope into a canonical schema or treat publication as Ratification.